Nitrate Contamination in Fresno, CA

What is the correlation between crop type in industrial agriculture (e.g., almonds and grapes) and nitrate contamination in domestic supply wells in Fresno County, CA?

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Background

California's Central Valley is one of the most productive agricultural regions in the world, generating over $50 billion annually and supplying a quarter of the U.S. supply of fruits, nuts, and vegetables[1]. This productivity stems from the region's fertile soils, warm climate, and reliance on groundwater[2]. However, these same conditions also promote nitrate leaching, especially where synthetic fertilizers are heavily applied[3].

Fresno County is the state's top agricultural producer, with over $8 billion in crop value in 2022[4]. High-nitrogen-use crops like almonds and grapes dominate its 1.88 million acres of farmland[5], contributing to nitrate contamination of groundwater. While this agricultural intensity sustains the regional economy, it also creates long-term water quality challenges. The intensive amount of agriculture, particularly the increasing rate of fertilizer application, over the last 50 years, has led to groundwater contamination with nitrate. Fresno County is home to over 1 million people, with 17.7% living below the poverty line, despite a median household income of $71,434[7]. This contamination disproportionately affects rural residents who rely on private domestic wells, which are largely unregulated. Of the 5,311 domestic wells in the county, only 490 are monitored for nitrate[8].

Methodology

Import Datasets - (LandIQ, GAMA Monitor Wells, US Census, Disadvantaged Communites)

Data Preparation - Select by Location of datasets intersecting with Fresno County boundary line from US Census dataset. Filter and clean null values data, and remove duplicates.

Buffer-Based Exposure Assessment (LandIQ of Grapes and Almonds, Monitor Wells, DAC)

A 4.8 km buffer was created around each monitor well to approximate agricultural influence. Almond and grape fields were clipped to these buffers, and **percent crop coverage** was calculated per buffer zone.

Statistical Modeling: An **Ordinary Least Squares (OLS)** regression was conducted to examine the correlation between nitrate levels and percent crop coverage, controlling for well depth. Residuals were tested using **Global Moran's I**, confirming spatial independence.

Spatial Prediction: **Kriging interpolation** was applied to predict nitrate concentrations across the county based on monitor well data. The raster output was **clipped to the Fresno County boundary** to avoid extrapolation beyond the observation extent. Domestic wells were overlaid on the Kriging surface to assess predicted nitrate exposure in unmonitored areas

Exposure Mapping: The Kriging surface was overlaid with **domestic wells** to assess potential exposure in areas lacking direct monitoring. Final outputs included nitrate prediction maps and spatial overlays identifying regions of concern.

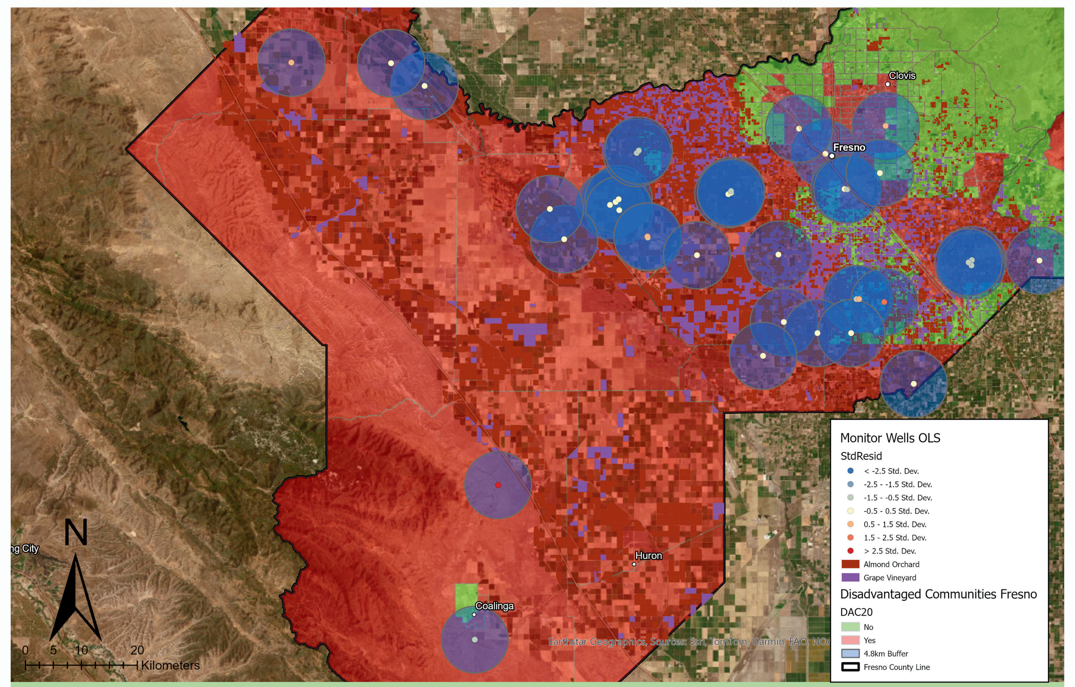
Data Sources

Data Layer	Source Type
Crop type (Almonds & Grapes)	LandIQ. (2023). Land use mapping. Retrieved March 13, 2025.
Domestic and Monitor Wells	California State Water Resources Control Board. (2025). GAMA groundwater information system. Retrieved March 13, 2025.
California Disadvantaged Communities (DAC)	California Office of Environmental Health Hazard Assessment. (2022). SB 535 disadvantaged communities. California State Geoportal. Retrieved March 15, 2025.
Census Tract	Federal Geospatial Data Committee. (2021, June 29). U.S. Census Blocks. ArcGIS Hub. Retrieved March 13, 2025.

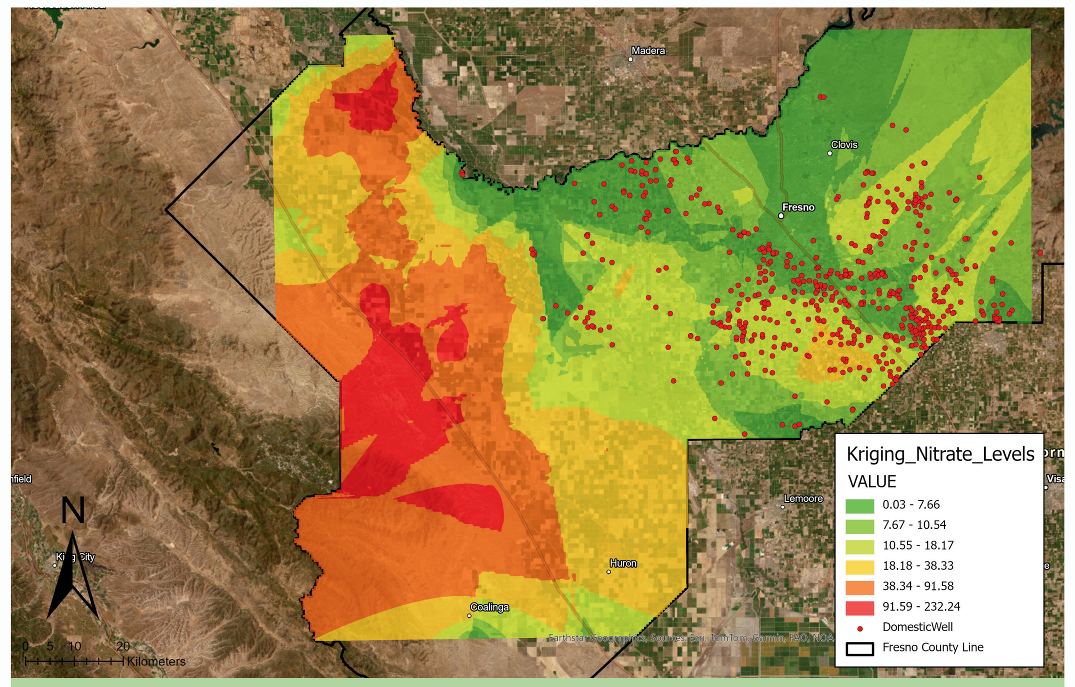
Results

To evaluate how crop type correlates with nitrate contamination in domestic water supplies, we ran an Ordinary Least Squares (OLS) regression using nitrate concentrations from 39 post-2014 monitor wells in Fresno County. Explanatory variables included percent almond coverage, percent grape coverage, and well depth, based on a 4.8 km buffer around each well.

The model produced a modest adjusted R^2 of 0.083, indicating that ~8% of nitrate variability is explained by the included variables. Despite this, the model was statistically valid with no multicollinearity since all Variance Inflation Factors (VIFs) are well below 7.5 (all VIFs < 2.5) and spatially independent residuals, as confirmed by Global Moran's I ($z = -1.69$, $p = 0.096$). These findings informed the broader analysis of monitoring gaps and infrastructure equity across the region.



Ordinary Least Squares (OLS) of Crop Type Density and Monitor Well Nitrate Contamination



Interpolated Nitrate Contamination in Fresno, CA

Table 1. Results of Ordinary Least Square Model

Variable	Coefficient	Robust p-value	Interpretation
Well Depth (ft)	-0.0549	0.0187	deeper wells → lower nitrate
% Almond Coverage	-0.5015	0.0694	more almond fields → marginal decrease in nitrate levels
% Grape Coverage	0.2152	0.6716	No significant relationship

While almond and grape coverage were hypothesized to influence nitrate levels, neither variable showed a statistically significant relationship in this model, suggesting that other unmeasured factors may be more explanatory.

Table 2. Moran's I Spatial Autocorrelation

Metric	Value	Interpretation
Moran's I	-0.206	Slight spatial dispersion
p-value	0.096	Not statistically significant
z-score	-1.69	Near edge of 90% confidence interval

The Moran's I result (-0.206, p = 0.096) indicates that model residuals are not significantly clustered, satisfying spatial independence assumptions.

Conclusions

This study examines the relationship between nitrogen-intensive grape and almond crops and nitrate contamination in Fresno County's domestic groundwater. Analyzing post-2014 monitored well data, we found that well depth significantly predicts nitrate levels, while almond crop coverage exhibits a marginal inverse relationship. This indicates complex interactions between land use and hydrology. Despite hypothesized influences from almond and grape coverage, neither showed a statistically significant correlation with nitrate concentrations, suggesting other unmeasured factors may play a larger role.

Over 90% of domestic wells are not regularly monitored, disproportionately impacting disadvantaged communities (DACs) that rely on these sources. Our regression modeling and kriging tools support spatial analysis, highlighting areas of nitrate risk overlapping with DACs lacking infrastructure.

Although modest, the model's explanatory power met statistical assumptions and revealed coherent spatial patterns. This underscores the need for expanded monitoring in DACs and suggests that trends in crop exposure can inform water vulnerability assessments.

Literature Cited

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Acknowledgments

Melissa Cragen - Agricultural Commissioner of Fresno County

Richard Nelson - California Water Resources Control Board, GAMA Unit

Kelly Garvey - PhD candidate at UCSB specializing in groundwater well trends

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